Overfitting

1 The Generalization Framework

Let the true data-generating distribution be $\mathcal{D}$ over an input space $\mathcal{X}$ and an output space $\mathcal{Y}$. Let the hypothesis class (the set of all possible neural networks with a fixed architecture) be $\mathcal{H}$.

A learning algorithm receives a finite **training sample** of size n, drawn independently and identically (i.i.d.) from $\mathcal{D}$:

$$S = \{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\} \sim \mathcal{D}^n$$

Let $h \in \mathcal{H}$ be a specific model (a specific set of weights). We define the **loss function** as $\ell(h(x), y)$.

We have two distinct error measures:

1. **Empirical Risk (Training Error):** The average loss on the training set S.

$$\hat{R}_S(h) = \frac{1}{n} \sum_{i=1}^{n} \ell(h(x_i), y_i)$$

2. **True Risk (Generalization Error):** The expected loss over the entire distribution $\mathcal{D}$ (unseen data).

$$R(h) = \mathbb{E}_{(x,y)\sim\mathcal{D}}[\ell(h(x), y)]$$

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2 Mathematical Definition of Overfitting

Let $\mathcal{A}(S)$ be the learning algorithm (e.g., stochastic gradient descent) that outputs a hypothesis $h_S \in \mathcal{H}$ trained on the sample S.

Overfitting occurs when the empirical risk is significantly lower than the true risk:

$$\hat{R}_S(h_S) \ll R(h_S)$$

Or equivalently, the **generalization gap** (or generalization error) is large:

$$\text{Gap}(h_S) = R(h_S) - \hat{R}_S(h_S) \gg 0$$

The Mechanism (How it happens): The learning algorithm minimizes the empirical risk. As model capacity increases, the algorithm finds a hypothesis h_S that perfectly interpolates the training data (so $\hat{R}_S \to 0$), but in doing so, it fits the *aleatoric noise* (random variance) present in the sample S. Because this noise does not exist in the true distribution $\mathcal{D}$, the true risk $R(h_S)$ remains high.

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3 The Bias-Variance Decomposition

To see this algebraically, let us look at the expected squared error for a regression problem. For a test point x, let $y = f(x) + \epsilon$, where $\mathbb{E}[\epsilon] = 0$ and $\text{Var}(\epsilon) = \sigma^2$ (the irreducible noise).

The expected mean squared error at x over all possible training sets is decomposed as:

$$\mathbb{E}_S\left[(y - h_S(x))^2\right] = \underbrace{(\mathbb{E}_S[h_S(x)] - f(x))^2}_{\text{Bias}^2} + \underbrace{\mathbb{E}_S\left[(h_S(x) - \mathbb{E}_S[h_S(x)])^2\right]}_{\text{Variance}} + \underbrace{\sigma^2}_{\text{Noise}}$$

- **Bias:** Error from erroneous assumptions in the learning algorithm (e.g., a linear model trying to fit a parabola). High bias causes **underfitting**.
- **Variance:** Error from sensitivity to small fluctuations in the training set. A model with high variance pays too much attention to the training data (including the noise ϵ).

Overfitting is the regime of High Variance: The model's output $h_S(x)$ changes drastically depending on which specific training set S it saw, meaning the Variance term dominates the error.

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4 The Capacity / Approximation-Estimation Tradeoff

Let $\mathcal{H}$ be the hypothesis class. The expected risk of the best model in $\mathcal{H}$ can be bounded by:

$$R(h_S) \leq \underbrace{R(h^*)}_{\text{Approximation Error}} + \underbrace{O\left(\sqrt{\frac{\text{Complexity}(\mathcal{H})}{n}}\right)}_{\text{Estimation Error}}$$

where $h^* = \arg\min_{h \in \mathcal{H}} R(h)$ is the best possible model in $\mathcal{H}$, and $\text{Complexity}(\mathcal{H})$ is a measure of the model's capacity (e.g., VC-dimension, Rademacher complexity, or number of parameters).

- If we restrict $\mathcal{H}$ (low complexity), the Estimation Error is small, but the Approximation Error is large (**Underfitting**).
- If we make $\mathcal{H}$ extremely complex (e.g., a massive deep neural network), the Approximation Error goes to zero. However, the Estimation Error grows because the term $\sqrt{\text{Complexity}(\mathcal{H})/n}$ increases. The algorithm can now find an h_S that minimizes $\hat{R}_S$ but is far from h^* in terms of true risk (**Overfitting**).